



Shape of Sound

A method that enhances emotion in voice messages

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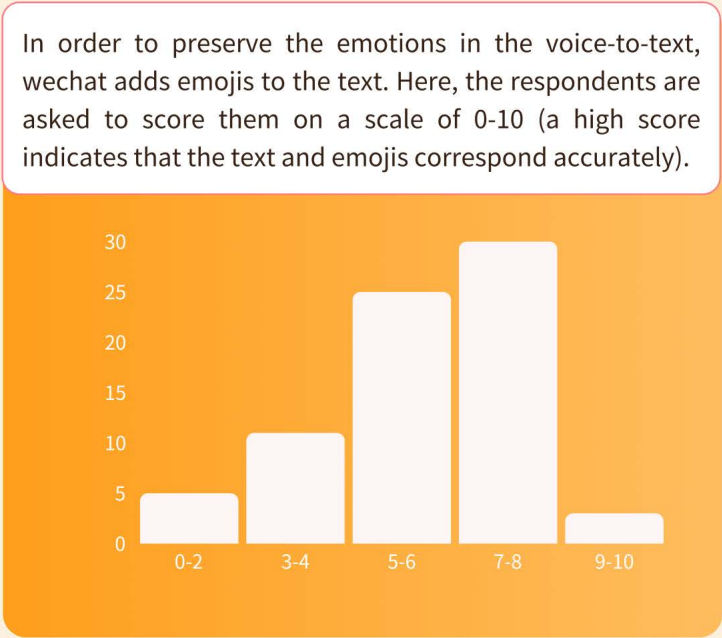
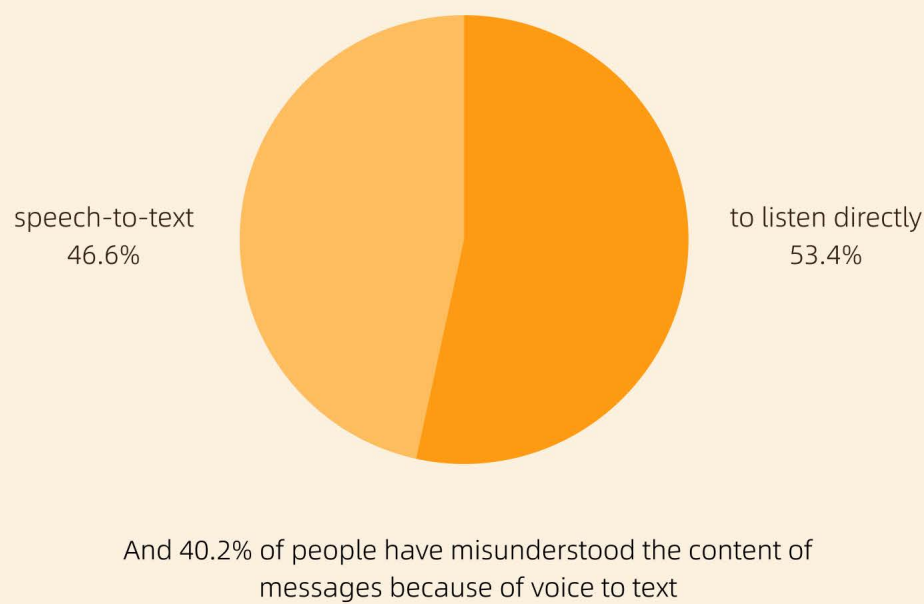
Using the existing model, we propose a method that can reduce the emotional loss in the process of speech-to-text by encoding the voice emotion into emoji, which can meet the needs of users to a certain extent, and has acceptable text accuracy and emotion enhancement scores.

INTRODUCTION

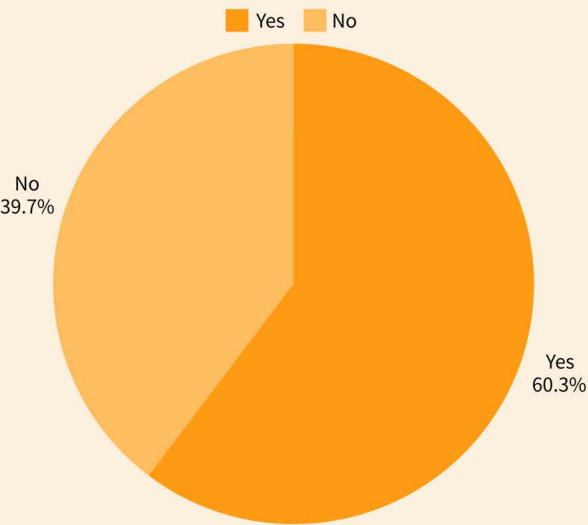
In modern life, voice information provides a richer medium for emotional expression, but sometimes it is inconvenient for people to listen to voice information, so the speech-to-text function comes into being. The existing speech-to-text function will remove the features such as pitch and sound size in the process of text conversion, and the disappearance of these features will make the rich emotion in the speech media disappear. We hope to design products that can communicate these features in a visual form and better preserve the emotional expression of the speech-to-text process.



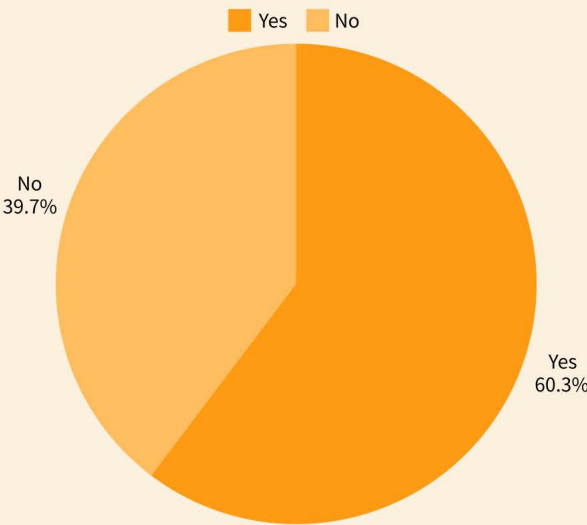
1. When you receive a voice message, are you more likely to listen directly to voice/speech-to-text?
2. Have you ever misunderstood the content of someone's message because of voice to text?
3. Do you think the expressions added by wechat voice to text match the emotions contained in the content?



4. If users would like to experience the text size changing with the volume after voice to text?



5. Do users think the full-screen motion effect after sending a message is an interesting design?



We **can** enhance the expression of emotions in voice messages by combining visual representations

PERSONA



Name: A

Information: Age 29, freelance

Demand: I don't want to waste time listening to the voice, but I also hope that the text can feel the emotion, and I feel that the text transferred from the voice is a little cold.

Pain point: Existing speech-to-text does not reflect much emotion.



Name: B

Information: Age 50, migrant worker

Demand: Meetings or other working hours are not convenient to listen to the voice, and when the text is changed, it is not urgent.

Pain points: Feeling no emotions can lead to mistakes at work.



Name: C

Information: Age 16, student

Demand: The voice is too lazy to listen, it sounds too slow. And some people are afraid to listen to people's voices.

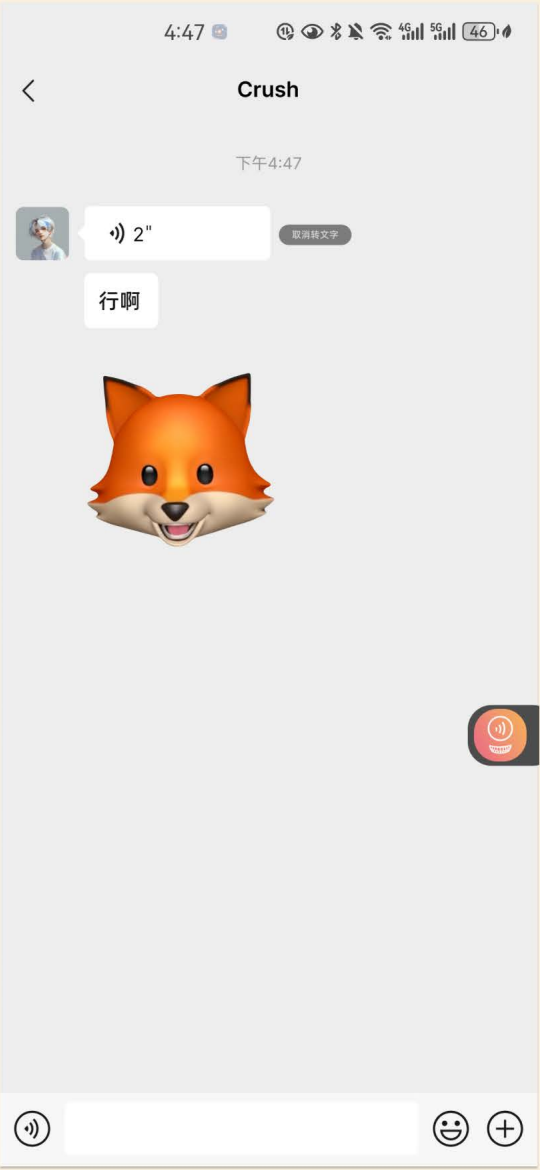
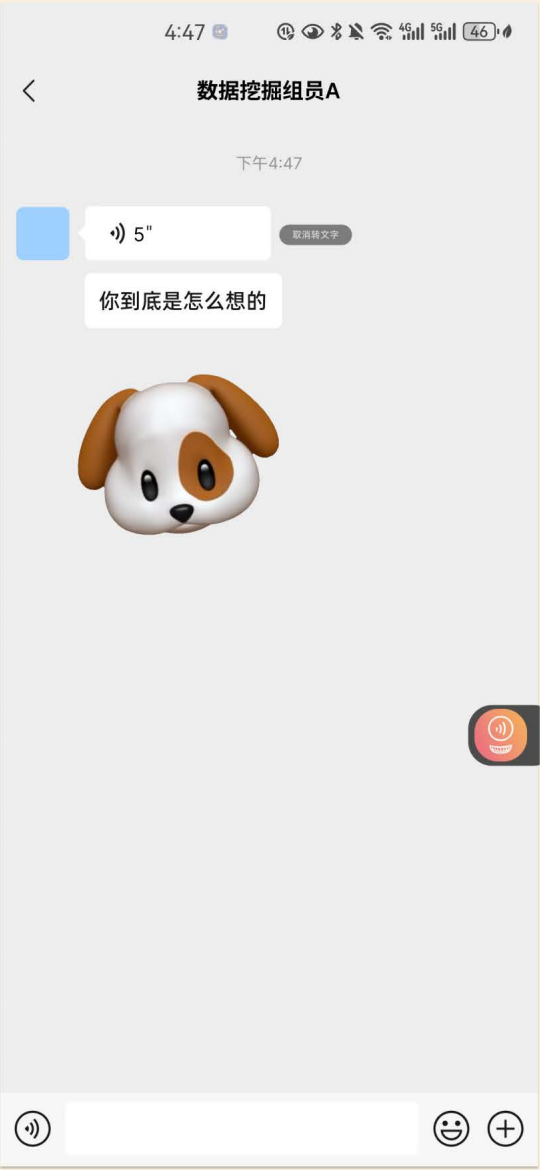
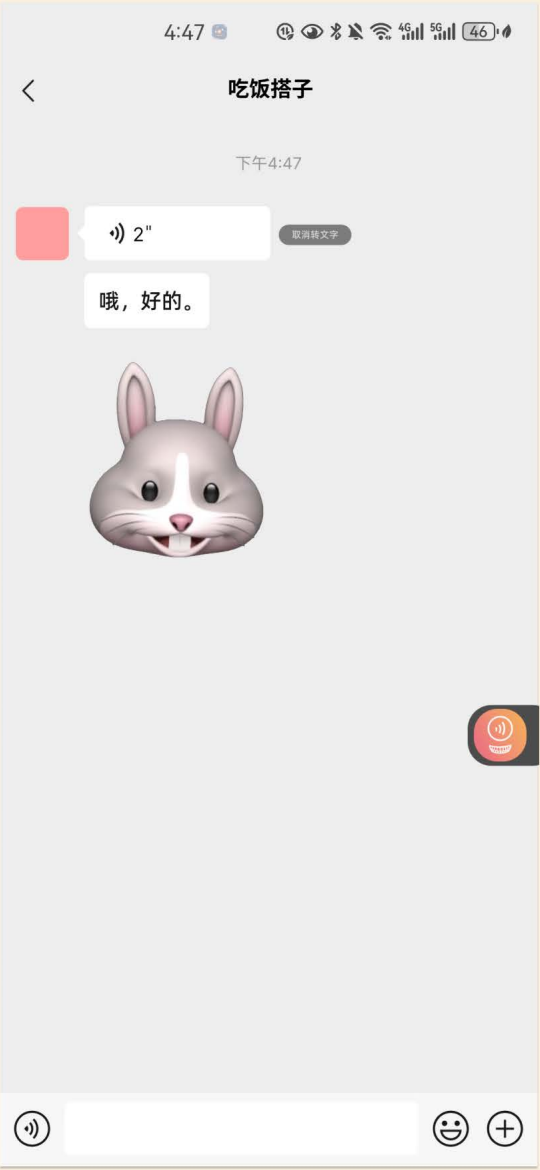
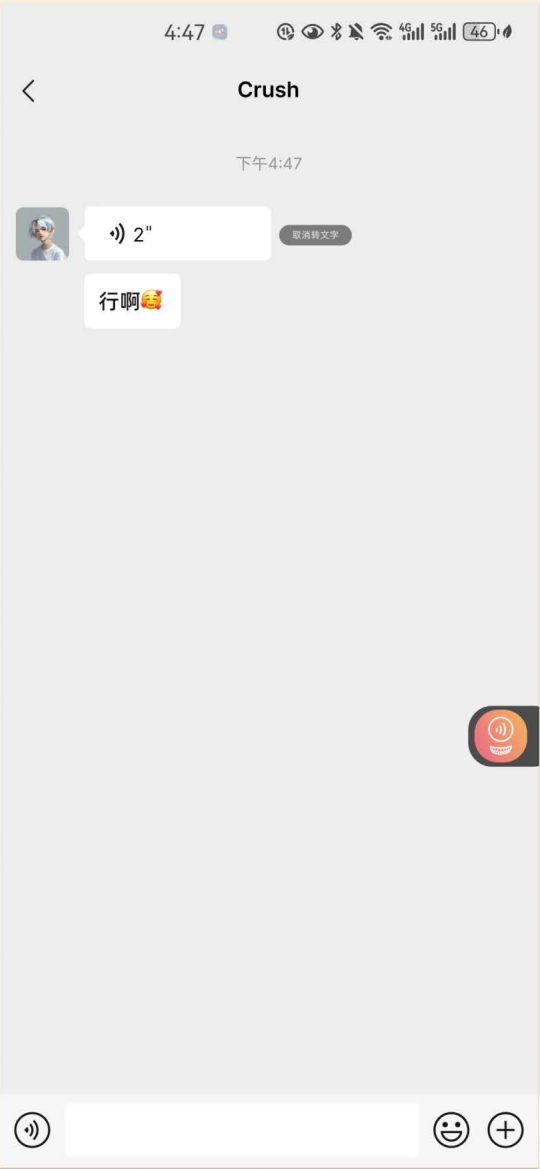
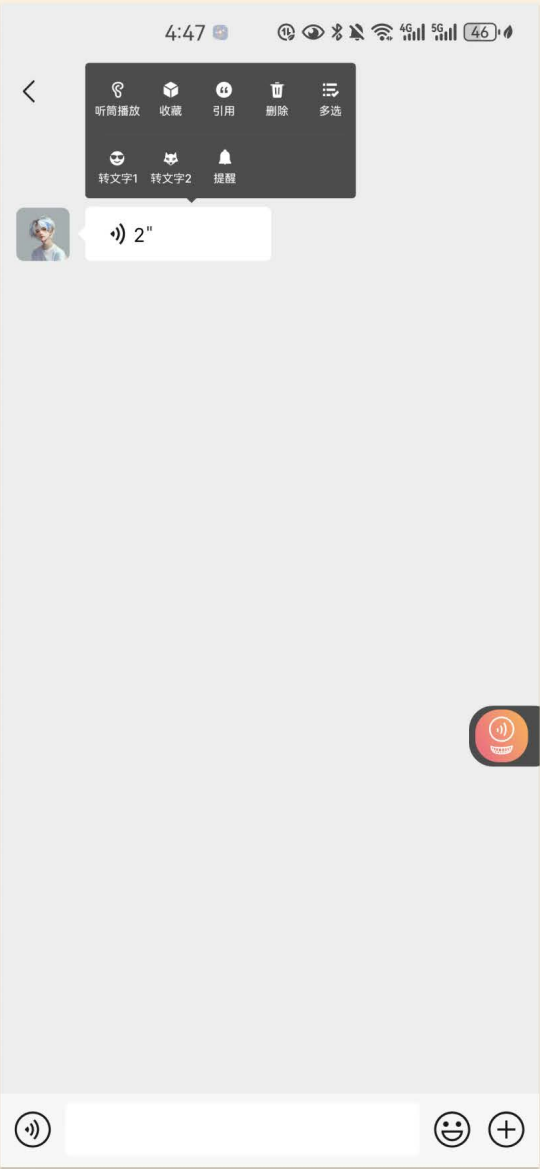
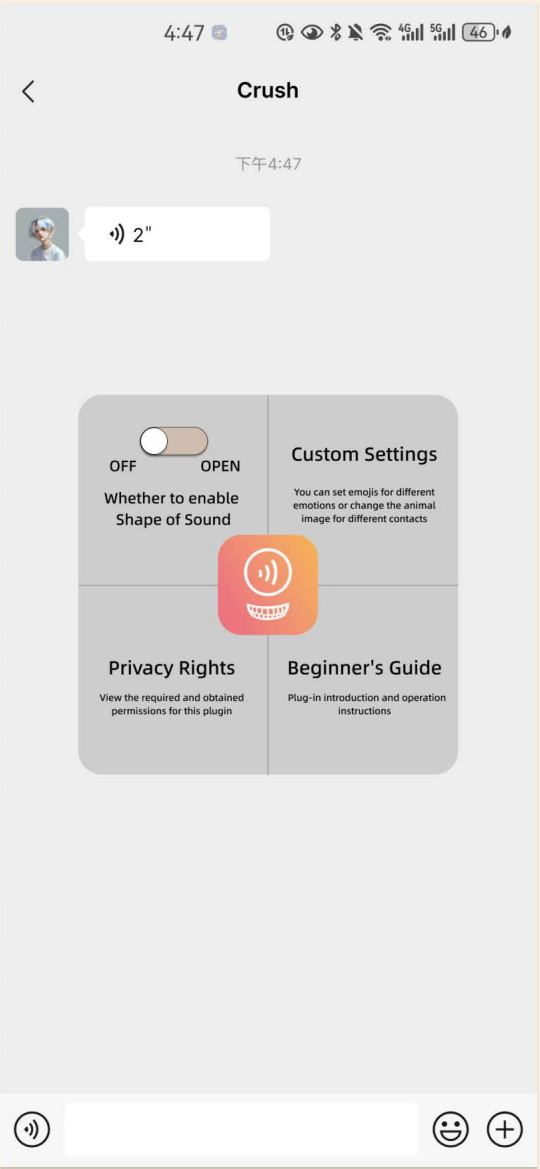
Pain point: Existing speech-to-text has led to misunderstandings.

METHOD DETERMINATION



We selected five ways to enhance emotional expression in speech-to-text, and surveyed 41 users with a questionnaire, and finally concluded that **adding emojis** and **virtual character** are two best way for users to understand the sender's emotions.

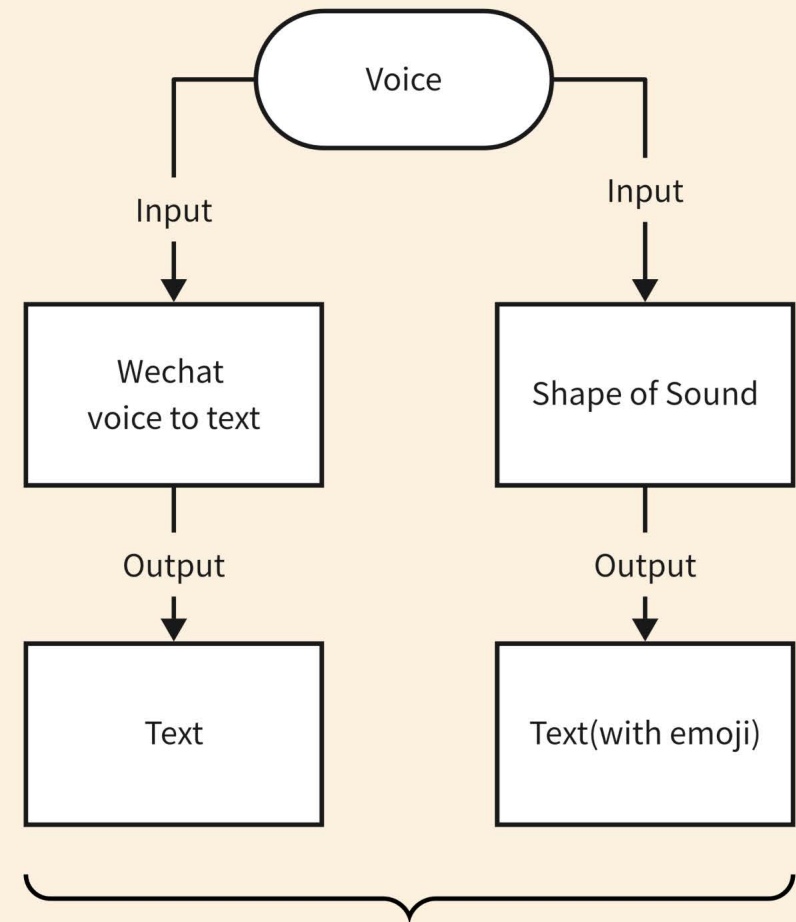
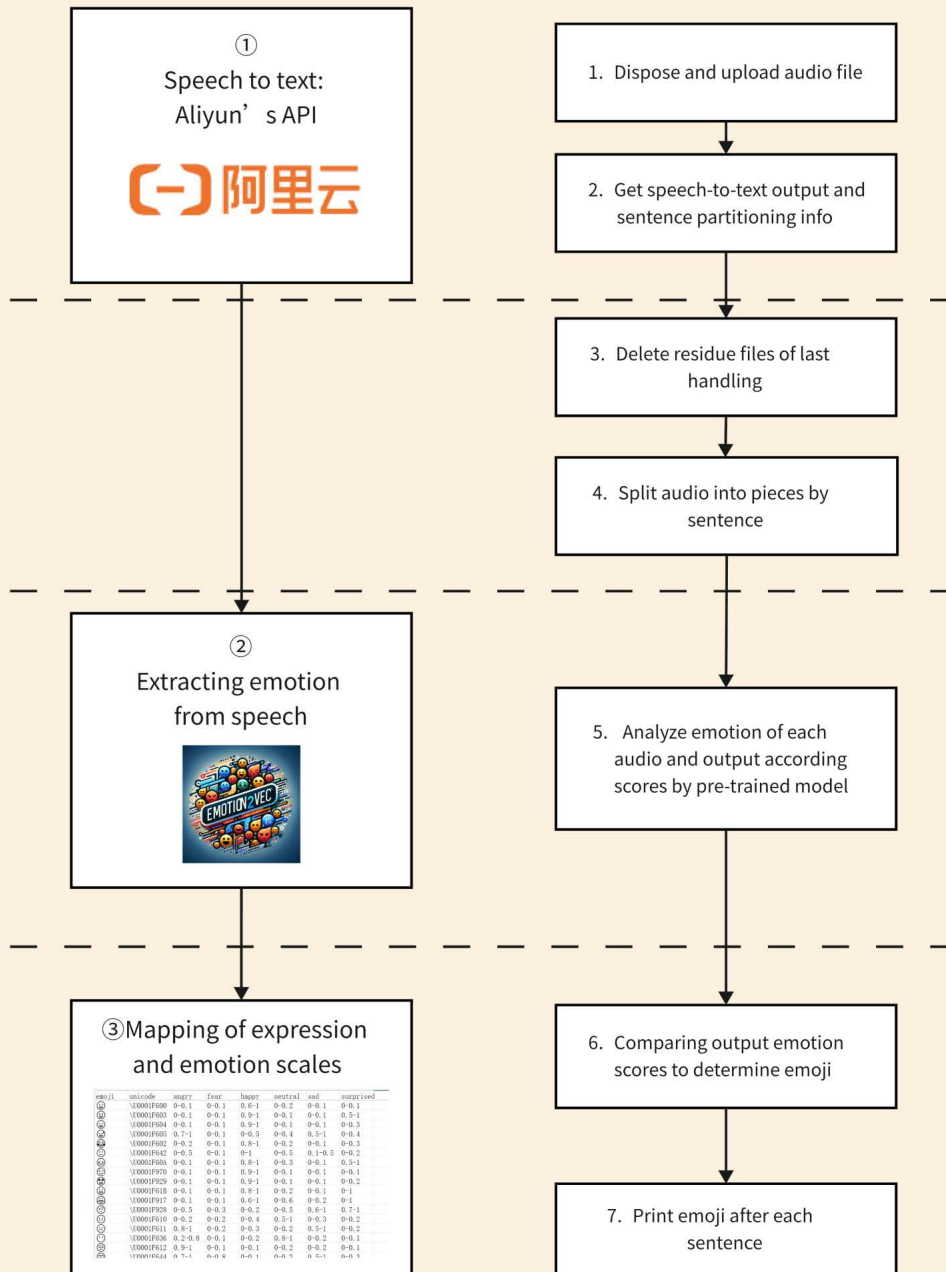
PROTOTYPE



We designed two prototypes to collect user feedback.

<https://www.figma.com/proto/uCM8TL433ojyrGTMhzwblZ/HCI-%E4%BA%BA%E6%9C%BA%E4%BA%A4%E4%BA%92?node-id=43-559&t=Sb5xMHXc4NLep6Nq-1&scaling=scale-down&page-id=0%3A1&starting-point-node-id=43%3A559>

PIPELINE AND EVALUATION



Evaluation: (Likert scale)

1.Accuracy of speech to text

2.Semantic comprehension difficulty of the text itself
(irony and similar dimensions affect this)

3.Affective enhancement

RESULT

We invited 15 participants (M=6, F=9) each to send 3 voice messages larger than 5 seconds, input the voice messages into wechat speech-to-text system and our system respectively, and then ask the participants to rate the accuracy of text translation, the difficulty of semantic understanding of sentences (which will higher if irony is in sentences) and the degree of emotional enhancement respectively. Scoring was performed using a five-point Likert scale.

Average score of accuracy average score: SoS: 4.61 wechat: 4.53 Errors mainly focus on mixed Chinese and English translation Emotion enhancement average score: SoS: 3.76 Wechat: 1.72 (Only 20% of the voices triggered the emoji, and those that did not were scored as 1 point) The average score of semantic comprehension difficulty: 2.3 Almost no participants uploaded meaningful and incomprehensible speech.

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ADDITIONAL STATEMENT

It was the spring of 2024 the human-computer interaction course project, I as a team leader in the project and responsible for all the questionnaire design and prototyping, other documents about the project at <https://github.com/guaiwu012/CS160-HCI>

In this project, we experienced a series of classical methods of human-computer interaction such as formative research, usability research, and user research, but we could not really make them into products due to time constraints. Coincidentally, at the beginning of our research on wechat, there were only three emojis showed in its text transfer function, and the trigger frequency was very low. When we finished our research, we found that it improved the trigger sensitivity, and basically every voice containing special emotions could be translated into text with emojis. We follow the product development closely, which is exciting!